NHAI Makhu Town EPC Agreement

Benchmark contract BM1 of 4 · CE725 Team 32, Topic 12 (AI Contract Analyser for Financial Risk Clauses) · prepared 22 October 2026 by the Data Lead

1. What this document is

Contract	Draft Engineering, Procurement and Construction (EPC) agreement with schedules, July 2024: widening and reconstruction of the existing two-lane carriageway to four lanes with paved shoulder in Makhu Town, km 170+000 to 171+900, including an ROB at km 171+387, NH-54, Punjab.
Issuer	National Highways Authority of India (under the Ministry of Road Transport and Highways). Contractor and Contract Price are blank placeholders in the draft.
Why it is the best of the four	It is the only one with all six clause types stated with specific numbers, in a long realistic document (250 pages) with a clean text layer. It also holds the classic traps: a rate and cap that disagree, layered guarantees, and formulas that are images.
Raw file	BM1_NHAI_Makhu_Town_EPC_Agreement.pdf · 250 pages · 3,389,664 bytes · free to download, no login
Source address	https://nhai.gov.in/nhai/sites/default/files/2020/EPC_Agreement_Schedules-Makhu_Town.pdf (downloaded 22 October 2026)
SHA-256 fingerprint	`90d6c58c9f2d2ca9ab8d28892134a3fc77525f78874d52524f60a41c52ba7d5e`
Integrity check	A fresh download from the source address matched the file byte for byte. The file in the Benchmark Contracts folder is unchanged.
Text layer	All 250 pages have readable text. Five pages also carry images, including the price-adjustment formulas (PDF pages 109 to 111).

2. What it says about the six clause types

Clause type	What the contract says	Clause	PDF page
Payment	Stage payments, one statement a month by the 7th. Interim certificate within 15 days. Paid no later than 30 days from the statement. Late payment interest: Bank Rate + 3% a year.	19.3 to 19.5, 19.9	105 to 108
Retention	6% deducted from every payment, but total capped at 5% of Contract Price. Balance refunded within 15 days of the Completion Certificate.	7.5	50
Advance	10% of Contract Price, interest-bearing at Bank Rate + 3%, in two instalments of 5%. Each needs an e-bank guarantee for 110%. Recovered at 15% of each stage payment, fully repaid before 80% of the construction period. Extra equipment advance up to 5%.	19.2	102 to 104

Clause type	What the contract says	Clause	PDF page
Delay damages	If a project milestone or the Scheduled Completion Date is not achieved within 30 days of the date in Schedule J, damages of 0.05% of Contract Price per day run from the Schedule J date. Total capped at 10%; beyond that the Authority may terminate. Refunded if the project finishes within the scheduled date including extensions.	10.3	64 to 65
Price escalation	Index formula applied in each interim certificate (roads, bridges, and a quarterly formula for maintenance). No threshold or cap. Payable only for stages billed within 30 days of the milestone or completion date.	19.10, 19.11	108 to 111
Guarantees	Performance Security 3% of Bid Price, valid 60 days after the defects liability period. Additional security for bids more than 20% below estimate, capped at 3%. Advance guarantees at 110%.	7.1, 7.2, 19.2	48, 103

Page numbers are positions in the PDF file, not the printed page numbers. Full quotes and every question are in section 5.

3. Arithmetic from the terms (ours, not in the contract)

Days of delay to hit the 10% damages cap	10% / 0.05% a day = 200 days of delay
Cumulative payments at which retention reaches its cap	5% / 6% = 83.3% of Contract Price paid
Cumulative payments at which the 10% advance is fully recovered at 15% a time	10% / 15% = 66.7% of Contract Price paid (the contract also demands full repayment before 80% of the construction period)
Advance paid out	5% at the first request, a further 5% after day 60 from the Appointed Date and a utilisation certificate

No rupee amounts can be worked out because the Contract Price is blank. The late-payment and advance interest need the current RBI Bank Rate, which is not in the contract.

4. Where an AI tool is likely to go wrong

Rate and cap that disagree	Retention is 6% of every payment but capped at 5% of Contract Price. A tool that reports only one number is wrong.
Two dates for Milestone I	Schedule J says day 145. The worked example in clause 10.3(ii) (PDF page 65) says day 180. The key follows Schedule J.
Formulas are images	The price-adjustment equations are pictures in the PDF. A text-only tool sees only the variable definitions and may invent the algebra. We read them from the page image.
Three different interest rates	Late payment by the Authority and the advance both carry Bank Rate + 3%. If the advance is recovered by encashing the guarantee, it is deemed to carry Bank Rate + 5% (PDF page 104). If the Authority returns a performance security late, it pays a fixed 9% a year (PDF page 50).

Two separate 110% guarantees	Each 5% advance instalment has its own guarantee. A tool may merge them into one 10% guarantee.
'Damages', not 'liquidated damages'	The contract calls delay payments 'Damages' and says they are a genuine pre-estimate of loss, 'not by way of penalty or liquidated damages' (clause 1.2(w), PDF page 26). A tool that searches for 'liquidated damages' will miss clause 10.3 or misread this sentence as saying there are none. The Damages are capped at 10% and refunded if the project finishes within the scheduled date.
Blank placeholders	The contractor's name and the Contract Price are blank in this draft, so rupee amounts cannot be worked out from the document.

5. The answer key: 52 questions

Statuses: Found (stated), Not stated (nothing said), Not applicable (clause switched off), Points elsewhere (defers to another document). Questions come from Contract_Extraction_Questionnaire_v0.1.md.

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
0.1	Contract type, form and version	Found	NHAI draft EPC agreement for four-laning of Makhu Town, NH-54 Punjab; cover dated July 2024	Cover page	1	Draft Engineering Procurement and Construction (EPC) Agreement For Widening and Reconstruction of existing Two Lane Carriageway to Four Lane with Paved Shoulder in Makhu Town
0.2	Employer and contractor	Found	Employer: President of India through MoRTH, represented by Chairman NHAI (Chandigarh regional office); Contractor: not named (placeholder)	Parties clause	14	The President of India through the Ministry of Road Transport & Highways, Government of India represented by: The Chairman, National Highways Authority of India, Regional office, Chandigarh
0.3	Contract value / price	Points Elsewhere	Lump sum Contract Price left blank (INR ...) to be filled with the accepted bid price	Cl. 19.1(i)	102	the lump sum price accepted by the Authority in consideration of the obligations specified in this Agreement for an amount of INR … … … … (INR … … … … … … … … …) (the "Contract Price")
0.4	Completion period and start trigger	Found	730 days from the Appointed Date	Cl. 10.3(i)	64	The 730th (seven hundred and thirtieth) day from the Appointed Date shall be the scheduled completion date
0.5	Defects liability / maintenance period	Found	5 years (flexible pavement); 10 years (rigid pavement, perpetual-design flexible, stand-alone structures), from completion	Cl. 17.1(i)	95	(a) 5 (five) years from the date of completion in case of a road being constructed with flexible pavement; (b) 10 (ten) years from the date of completion in case of road being constructed with rigid pavement;

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
0.6	Milestones	Found	Milestone I day 145 (statements >=10% of price); II day 320 (>=35%, all bridges started); III day 520 (>=70%); Scheduled Completion day 730	Sch. J paras 2-5	219	Project Milestone-I shall occur on the date falling on the 145th (one hundred and forty fifth first) day from the Appointed Date
1.1	Billing basis and frequency	Found	Stage payments for completed stages valued per Schedule-H; contractor submits one Stage Payment Statement a month by the 7th	Cl. 19.3 and 19.4	105	The Contractor shall submit a statement (the “Stage Payment Statement”), in 3 (three) copies, by the 7th (seventh) day of the month
1.2	Days to certify a bill	Found	Authority's Engineer issues the Interim Payment Certificate within 15 days of receiving the statement; within 10 days he recommends a 90% part payment	Cl. 19.5(i) and (ii)	105	Within 15 (fifteen) days of the receipt of the Stage Payment Statement referred to in Clause 19.4, the Authority’s Engineer shall determine and shall deliver
1.3	Days to pay after certification	Found	Payment no later than 30 days from submission of the Stage Payment Statement; if no IPC in 30 days the Authority pays the amount claimed and adjusts later	Cl. 19.9(i)(a)	107	payment shall be made no later than 30 (thirty) days from the date of submission of the Stage Payment Statement by the Contractor
1.4	Interest on delayed payment	Found	Bank Rate + 3% per annum on all unpaid sums	Cl. 19.9(ii)	108	the Authority shall be liable to pay to the Contractor interest @ Bank Rate + 3% per annum, on all sums remaining unpaid
1.5	Deductions from each bill	Found	Retention 6% of every payment (cap 5% of Contract Price); advance recovery 15% of each Stage Payment Statement; withholding for unperformed or defective work (19.5(iv))	Cl. 7.5(i); 19.2(vi); 19.5(iv)	50	deduct 6% (six per cent) thereof as guarantee money for performance of the obligations of the Contractor during the Construction Period
1.6	Minimum bill or cap before certification	Found	No minimum bill value stated; before the IPC only 90% of the amount broadly determined is released as part payment	Cl. 19.5(i)	105	recommend the release of 90 (ninety) percent of the amount so determined as part payment against the Stage Payment Statement, pending issue of the Interim Payment Certificate
1.7	Final bill and final payment	Found	Final Payment Statement within 60 days of Completion Certificate; Final Payment Certificate within 30 days of statement and discharge; payment per 19.9	Cl. 19.13(i); 19.15; 19.9	111	Within 60 (sixty) days after receiving the Completion Certificate under Clause 12.2, the Contractor shall submit
1.8	Pay-when-paid or conditional payment	Not Stated	None found			

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
2.1	Retention deducted?	Found	Yes	Cl. 7.5(i)	50	From every payment for Works due to the Contractor in accordance with the provisions of Clause 19.5, the Authority shall deduct 6% (six per cent) thereof
2.2	Rate and base	Found	6% of every payment for Works (stage payments under Cl. 19.5)	Cl. 7.5(i)	50	deduct 6% (six per cent) thereof as guarantee money for performance of the obligations of the Contractor during the Construction Period
2.3	Cap on total retention	Found	5% of Contract Price	Cl. 7.5(i)	50	subject to the condition that the maximum amount of Retention Money shall not exceed 5% (five per cent) of the Contract Price.
2.4	Bid security / earnest money counted against retention	Found	No link to retention: Bid Security stays in force until substituted by the Performance Security; it is not converted into retention money	1.1 definition of Bid Security; 7.1(iv)	17	"Bid Security" means the bid security provided by the Contractor to the Authority in accordance with the Request for Proposal, and which is to remain in force until substituted by the Performance Security;
2.5	When retention is released	Found	Balance refunded within 15 days of the Completion Certificate (after set-off of appropriations)	Cl. 7.5(iii)	50	Within 15 (fifteen) days of the date of issue of the Completion Certificate, the Authority shall refund the balance of Retention Money remaining with the Authority
2.6	Retention replaceable by bank guarantee	Not Stated	No substitution mechanism in this agreement			
2.7	Interest on retention	Not Stated	None stated			
3.1	Mobilisation advance given?	Found	Yes, interest-bearing	Cl. 19.2(i)	102	The Authority shall make an interest-bearing advance payment (the "Advance Payment") @ "Bank Rate3 + 3%", equal to 10 % (ten percent) of the Contract Price,
3.2	Amount and instalments	Found	10% of Contract Price in two instalments of 5% each	Cl. 19.2(i)	103	The Advance Payment for mobilisation expenses shall be made in two instalments each equal to 5% (five percent) of the Contract Price.
3.3	Interest-free or interest-bearing	Found	Interest-bearing at Bank Rate + 3% per annum	Cl. 19.2(i); 19.2(vii)	102	The Authority shall make an interest-bearing advance payment (the "Advance Payment") @ "Bank Rate3 + 3%", equal to 10 % (ten percent) of the Contract Price,

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
3.4	Recovery method	Found	15% deducted from each Stage Payment Statement from the first one; fully repaid before 80% of the scheduled construction period	Cl. 19.2(vi)	104	deductions shall be made at the rate of 15% (fifteen percent) of each Stage Payment Statement until such time as the advance payment has been repaid;
3.5	Security for the advance	Found	Irrevocable unconditional e-Bank Guarantee for 110% of each instalment, valid until full repayment	Cl. 19.2(iii) and (iv)	103	for an amount equivalent to 110% (one hundred and ten per cent) of such instalment
3.6	Plant and machinery or secured advance	Found	Additional advance on newly purchased key equipment up to 5% of Contract Price against e-BG; no advance on materials found	Cl. 19.2(ii)	103	The maximum of such advance shall be 5% (five per cent) of the Contract Price
3.7	Conditions to receive the advance	Found	Apply after Appointed Date with 110% e-BG; second instalment after 60 days from Appointed Date, with e-BG and utilisation certificate; paid within 15 days of request	Cl. 19.2(iii)-(v)	103	The Contractor may apply to the Authority for the first instalment of the Advance Payment at any time after the Appointed Date
4.1	Delay damages imposed?	Found	Yes (called "Damages")	Cl. 10.3(ii)	64	it shall pay Damages to the Authority of a sum calculated at the rate of 0.05% (zero point zero five percent) of the Contract Price for delay of each day
4.2	Rate and base	Found	0.05% of Contract Price for each day of delay, reckoned from the date in Schedule J. Payable if the milestone or the Scheduled Completion Date is not achieved within 30 days of that date	Cl. 10.3(ii)	64	In the event that the Contractor fails to achieve any Project Milestone or the Scheduled Completion Date within a period of 30 (thirty) days from the date set forth in Schedule-J
4.3	Cap on damages	Found	10% of Contract Price; above that the Contractor is in default and the Authority may terminate	Cl. 10.3(iii)	65	the total amount of Damages under Clause 10.3 (ii) shall not exceed 10% (ten percent) of the Contract Price
4.4	Completion only or milestones too	Found	Both each Project Milestone and the Scheduled Completion Date	Cl. 10.3(ii)	64	In the event that the Contractor fails to achieve any Project Milestone or the Scheduled Completion Date
4.5	How the date can be extended	Found	Time Extension for delayed Right of Way or clearances, Change of Scope, Force Majeure, Authority-caused delay; notice within 15 business days or the right is lost	Cl. 10.5(i)-(iii)	66	the Contractor shall be entitled to extension of time in the Project Completion Schedule (the "Time Extension") to the extent that completion of any Project Milestone is or will be delayed by any of the following, namely:

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
4.6	Pre-estimate of loss; proof of loss	Found	Damages are stated to be a genuine pre-estimate of loss and not a penalty; proof of loss not mentioned	Cl. 1.2(w)	26	mutually agreed genuine pre-estimated loss and damage likely to be suffered and incurred by the Party entitled to receive the same and are not by way of penalty or liquidated damages
4.7	Recovery from bills, retention or guarantee	Found	Appropriate from Performance Security and Retention Money on Contractor Default; recoverable from amounts due after 20 days notice	Cl. 7.3(i); 7.5(ii); 19.19	49	Upon occurrence of a Contractor's Default, the Authority shall, without prejudice to its other rights and remedies hereunder or in law, be entitled to encash and appropriate the relevant amounts from the Performance Security as Damages
4.8	Other damages or penalties	Found	0.01% of Bid Price per day for late security; maintenance non-compliance payment cuts (Sch. M); Authority pays damages for late site handover (8.3); bid ban if >90 days late on projects of Rs 300 crore or more; early completion bonus 0.03% a day, max 5%; also: lane-closure delay damages 0.1% of the monthly maintenance payment for each day per 250 m stretch (page 89); 20% of the cost as Damages for failure to rectify defects (page 96, Art. 17); 120% of the actual removal cost if the contractor fails to remove equipment and debris (page 82)	Cl. 7.1(iii); 8.3; 10.3(iv); 19.20; Sch. M	49	The Damages shall be the sum calculated at the rate of 0.01% (zero point zero one per cent) of the Bid Price offered by the Selected Bidder
5.1	Fixed price or price variation	Found	Price variation clause applies (index-based adjustment)	Cl. 19.10(iv)	108	The Contract Price shall be adjusted for increase or decrease in rates and price of labour, cement, steel, Plant, machinery and spares, bitumen, fuel and lubricants, and other material inputs
5.2	Formula	Found	Road and other works: VRW = 0.85 x RW x [PL(LI-L0)/L0 + PA(AI-A0)/A0 + PF(FI-F0)/F0 + PB(BI-B0)/B0 + PM(MI-M0)/M0 + PC(CI-C0)/C0 + PS(SI-S0)/S0]. Bridges and structures: VBR = 0.85 x BR x [same terms without the bitumen term PB]. Maintenance lump sum, adjusted quarterly: V = P x (WI - W0)/W0, using the all-commodities WPI	Cl. 19.10(iv)(d)	108	Price adjustment for changes in cost shall be paid in accordance with the following formulae:

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
5.3	Components and weights	Found	Labour 20/20/20/15/15%; cement 5/nil/20/15/15%; steel nil/nil/nil/15/20%; bitumen nil/15/nil/nil/nil%; fuel 10% each; other materials 50/40/35/30/25%; plant 15% each (five work types)	Cl. 19.10(iv)(e)	110	Labour (PL) 20% 20% 20% 15% 15% Cement (PC) 5% Nil 20% 15% 15% Steel (PS) Nil Nil Nil 15% 20% Bitumen (PB) Nil 15% Nil Nil Nil
5.4	Index and source	Found	WPI machinery; WPI cement (OPC); WPI mild steel long products; WPI all commodities; CPI-IW Bathinda (Labour Bureau); bitumen price at Bathinda (HMEL) refinery; HSD price at IOC pumps in Punjab	Cl. 19.10(iv)(d) definitions	110	SO = The WPI for Mild Steel –Long Products for the month of the Base Date. SI = The WPI for Mild Steel –Long Products for the month three months prior to the month to which the IPC relates.
5.5	Base date	Found	Last day of the calendar month at least 28 days before the Bid Due Date	Cl. 1.1 (Base Date)	17	means the last date of the calendar month, which precedes the Bid Due Date by at least 28 (twenty eight) days
5.6	Threshold or cap on escalation	Not Stated	No threshold or ceiling stated			
5.7	Escalation after original completion date	Found	Payable only for stages billed within 30 days of the applicable milestone or Scheduled Completion Date (including extensions)	Cl. 19.11	111	Price adjustment shall be due and payable only in respect of the stages of Works for which the Stage Payment Statement has been submitted by the Contractor no later than 30 (thirty) days
5.8	Escalation on employer-supplied materials	Not Stated	No such provision found			
6.1	Securities required	Found	Bid Security (RFP stage); Performance Security; Additional Performance Security for low bids; Retention Money; e-BG for each advance (mobilisation and equipment)	Cl. 7.1; 7.5; 19.2	48	Within 30 (thirty) days of receipt of Letter of Acceptance, the selected Bidder shall furnish to the Authority in the form of Insurance Surety Bond
6.2	Amount and base	Found	Performance Security 3% of Bid Price; Additional Performance Security up to 3% of Bid Price; advance e-BG 110% of each instalment; equipment advance up to 5% of Contract Price	Cl. 7.1(i); 19.2	48	for an amount equal to 3% (three percent) of its Bid Price
6.3	Accepted forms	Found	Insurance Surety Bond, Account Payee Demand Draft, Banker's Cheque, irrevocable e-Bank Guarantee; physical BG also acceptable till 31.12.2024	Cl. 7.1(i)(A) and footnote	48	Account Payee Demand Draft, Banker's Cheque or an irrevocable and unconditional e-Bank Guaranteeα from a Bank
6.4	Acceptable issuers	Found	A bank incorporated in India and recognised by the Reserve Bank of India	Cl. 1.1 (Bank)	17	a bank incorporated in India and recognized by the Reserve Bank of India

Q	Question	Status	Answer	Clause	PDF page	Quote from the contract
6.5	Validity and extension	Found	Performance Security valid until 60 days after the Defects Liability Period (may be furnished for 2 years and extended 2 months before expiry); Additional Performance Security valid until 28 days after Project Completion Date	Cl. 7.1(i)(B); 7.2	48	The Performance Security shall be valid until 60(sixty) days after the Defects Liability Period. The Additional Performance Security shall be valid until 28 (twenty eight) days after Project Completion Date.
6.6	Encashment grounds	Found	On Contractor Default; on termination for Contractor Default; advance guarantee if advance not recovered	Cl. 7.3(i); 23.6(i); 19.2(vi)(c)	49	Upon occurrence of a Contractor's Default, the Authority shall, without prejudice to its other rights and remedies hereunder or in law, be entitled to encash and appropriate the relevant amounts from the Performance Security as Damages
6.7	When released	Found	Performance Security within 60 days of the end of the Maintenance Period or Defects Liability Period, whichever is later; Additional Performance Security within 28 days of Completion Certificate	Cl. 7.4(i) and (ii)	50	The Authority shall return the Performance Security to the Contractor within 60 (sixty) days of the expiry of the Maintenance Period or the Defects Liability Period, whichever is later
6.8	Extra guarantee requirements	Found	Additional Performance Security if bid is more than 20% below estimate: 20% of the shortfall, capped at 3% of Bid Price	Cl. 7.1(i)(A)(a) and (b)	48	If the Bid Price offered by the Selected Bidder is lower than 20% of the Estimated Project Cost/ cost put to tender, the Additional Performance Security shall be calculated @20% of the difference

6. How the answers were checked

<table>
<tr><td>Check 1: text reader</td><td>All 47 quotes were found word for word on the stated page in text extracted by pypdf.</td></tr>
<tr><td>Check 2: second reader</td><td>The same 47 quotes were found on the stated page by a different PDF engine (PyMuPDF). The 5 answers that say nothing is stated carry no quote by design.</td></tr>
<tr><td>Check 3: blind re-read</td><td>A fresh reader who never saw the key answered all 52 questions. The two agreed on 48. The four differences:
<table>
<tr><th>Q</th><th>Difference</th><th>Outcome</th></tr>
<tr><td>0.1</td><td>Wording only</td><td>Key kept</td></tr>
<tr><td>1.5</td><td>Wording only (blind reader listed extra items)</td><td>Key kept</td></tr>
<tr><td>2.4</td><td>Blind reader found the Bid Security definition on page 17</td><td>Key changed from 'not stated' to 'found: no link to retention'</td></tr>
<tr><td>4.8</td><td>Blind reader listed more damages items</td><td>Three re-checked in the text and added; one could not be re-found and was left out</td></tr>
</table>
</td></tr>
<tr><td>Check 4: page images</td><td>PDF pages 109 and 111 were viewed as images to read the price-adjustment formulas, which are pictures in the PDF. They are now in answer 5.2. Weights and index definitions on those pages agree with the text answers.</td></tr>
</table>

What this does and does not prove. A quote found on the stated page proves the text exists there, not that the answer reads it correctly. Two text readers agreeing is also not proof. A person still has to compare each answer with its PDF page, especially the six main numbers: retention, advance, damages rate, damages cap, payment days and performance security.

7. Open items for this contract

- Contractor name and Contract Price are blank placeholders in the draft agreement (key rows 0.2 and 0.3).
- Schedules A to D and H (including the stage-payment schedule) do not appear in the extracted text. The percentages that decide how much is paid at each stage were not checked.
- Milestone I appears as day 145 in Schedule J and day 180 in the clause 10.3(ii) example. Confirm which governs.
- The agreement is a draft dated July 2024. Final signed versions of NHAI agreements may differ.
- A person must still compare each answer with its PDF page. Page numbers are positions in the file, not printed page numbers.

Companion files: Contract_Answer_Key_v2_verified.xlsx (all four contracts), Contract_Source_Register.xlsx (every source, labelled), Contract_Extraction_Questionnaire_v0.1.md (the 52 questions).